

IIN 3502 Distributed Network

Lab Sheet 04

Index No. – ICT/23/810

Name – A.K. Nipuna Darshana

1. What is Spanning Tree Protocol, and why is it needed in a switched network?

STP is a layer 2 protocol that prevents physical switching loops. It is needed because ethernet frames have no TTL field, if a loop exists, broadcast frames will circulate endlessly, causing catastrophic network failure.

2. What is a Layer 2 switching loop? Describe two problems it can cause.

Switches learn the same MAC address on multiple different ports

Broadcasts multiply and consume all available bandwidth

3. What is a Root Bridge, and how does STP elect one?

Root Bridge is the central reference point for the STP topology. STP elects it by exchanging BPDUs. The switch with the lowest Bridge ID is elected as the Root Bridge.

4. What is a Bridge ID? Explain the roles of bridge priority and MAC address.

Bridge ID is an 8-byte field consisting of a 2-byte Priority (default 32768) and a 6-byte MAC address.

5. What is the difference between a Root Port and a Designated Port?

- Root Port: The port on a non-root switch that has the lowest path cost to the Root Bridge. It always forwards traffic toward the Root.
- Designated Port: The port on a switch segment that has the lowest path cost to the Root Bridge. It forwards traffic away from the Root onto that specific segment.

6. What is an Alternate Port, and why might it be placed in a blocking state?

An Alternate Port is a non-designated port that is blocked by STP. It is placed in a blocking state to break the Layer 2 loop. It remains on standby and takes over if the active link fails.

7. What is STP path cost? How does link speed influence path selection?

Path cost is a value assigned to a link based on its bandwidth. STP calculates the root path cost by adding up the costs of all links to the Root Bridge

8. Why must VLANs 10 and 20 be considered separately when using PVST+?

Per-VLAN Spanning Tree Plus runs a separate STP instance for each VLAN. This allows for load balancing. For example, Switch 0 can be the Root for VLAN 10, while Switch 1 can be the Root for VLAN 20, utilizing different physical links for different VLAN traffic.

9. What happens when an active link fails? How does STP respond?

When a link fails, STP detects the loss of BPDUs. It triggers a topology change, recalculating the tree.

10. Why is a successful ping not sufficient proof that STP is working correctly?

A successful ping just means one specific IP packet made it through. It does not prove that STP has successfully blocked redundant paths for broadcast/multicast traffic. Only checking show spanning-tree proves loop prevention is active

11. What is the purpose of BPDUs?

Bridge Protocol Data Units (BPDUs) are messages exchanged between switches. They are used to elect the Root Bridge, determine the shortest path to the root, and detect network topology changes

12. What could happen if STP were disabled on all switches while the triangle topology remained connected?

The network would immediately experience a Broadcast Storm. Broadcast frames would loop endlessly between the three switches, driving CPU utilization to 100% and consuming all bandwidth, causing the network to completely crash